Modeling Electric Vehicle Efficiency and Performance with Machine Learning

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Executive Summary

The purpose of this project was to analyze how the technical specifications of electric vehicles (EVs) influence their efficiency and performance using the *Electric Vehicle Specs* dataset from Kaggle. Through clustering and predictive modeling, we identified key factors that distinguish EVs by performance and efficiency. Gaussian Mixture Model (GMM) clustering revealed distinct groups of EVs based on attributes like battery capacity, range, and speed. Linear regression analysis showed that efficiency is primarily driven by factors such as battery size, range, and towing capacity, while drivetrain type (AWD, FWD, RWD) had little effect. Logistic regression accurately predicted drivetrain categories with around 90% test accuracy, and a multiple linear regression model successfully predicted vehicle range with high reliability ($R^2 \approx 0.96$). Overall, the results demonstrate that machine learning can effectively uncover relationships between EV specifications and performance, offering insights that can guide manufacturers in optimizing design and help consumers make informed decisions about efficiency and range.

Introduction

In May of 2025, the International Energy Agency (IEA) found that electric vehicle sales occupied more than 20% of the market share in 2024, exceeding $17 million (IEA, 2025). This growth signals a change in consumer trends towards sustainable transportation. Several factors have contributed to this growth, including increased consumer awareness of environmental issues, government incentives and the regulatory mandates that aim to eliminate carbon emissions, and the declining cost of EV batteries, which makes large-scale production more economically viable (Fitzgerald, 2024).

Consumers must evaluate many attributes when purchasing an EV, including driving range, charging speed, and overall energy efficiency. This project aims to examine the technical attributes of EVs that contribute to higher energy efficiency and performance. Using *Electric Vehicle Specs* from Kaggle, we will use supervised and unsupervised machine learning techniques to uncover the relationships between specifications and performance outcomes.

Purpose and Scope

The purpose of this project is to explore how the technical features of EVs relate to their efficiency and performance. As EV adoption continues to grow around the world, understanding what factors make certain models more energy-efficient or longer-range can help manufacturers improve design and help consumers make better purchasing decisions. The dataset includes 478 EV models and 22 technical attributes such as battery capacity, torque, and efficiency, allowing for both numerical and categorical analysis.

This project uses machine learning to:

- Examine whether certain technical features (like battery capacity, number of cells, or torque) are related to each other and to overall vehicle performance.
- Identify which factors most influence an EV's efficiency, such as range, speed, and battery size.
- Build predictive models that estimate an EV's driving range based on measurable technical specifications.
- Use clustering methods to see if natural groupings of EVs exist, such as economy, mid-range, and luxury categories.

The scope of this project includes data cleaning, feature selection, clustering, and regression modeling on the *Electric Vehicle Specs* dataset from Kaggle. We used both unsupervised learning (for grouping similar EVs) and supervised learning (for predicting efficiency and range). Our goal is to uncover meaningful relationships between EV features and performance outcomes and to highlight how machine learning can support data-driven insights in sustainable transportation design.

Data Preparation and Preprocessing

Within the dataset, there are 22 attributes used to describe the different specifications of 478 electric vehicle models. These attributes include the brand, model, maximum speed, battery capacity, battery type, number of cells in the battery pack, motor torque, energy efficiency, driving range, acceleration capabilities, fast-charging rate, type of charging connector, towing capacity, cargo capacity, number of passenger seats, vehicle powertrain layout, market classification, physical dimensions of the vehicle in millimeters, vehicle body style type, and the reference link to the data source.

An initial inspection using `df.head()` and `df.info()` revealed that most variables were complete, with a few missing values in *number_of_cells*, *towing_capacity_kg*, and *torque_nm*. The dataset used consistent metric units, but some text fields contained non-numeric characters that required preprocessing.

Missing entries were removed using `df.dropna()`, resulting in a clean dataset of 264 rows and 22 columns. Quantitative variables such as top speed, torque, and efficiency were standardized to ensure consistent scaling for clustering and regression analyses.

Methodology and Analysis

3.1 Clustering Analysis: Relationship between Technical Qualities of the Batteries

Methodology

This section explores whether the technical specifications of electric vehicle batteries, like the capacity (kWh), number of cells, and efficiency, show a correlated relationship with performance measures like range and charging capabilities. We initially set the scales of quantitative features like the top speed, torque, range, acceleration, and energy efficiency. We then applied a Gaussian Mixture Model (GMM) across 2 to 20 clusters to uncover natural groupings. For both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), values were collected to compare the model fit and complexity.

Additionally, we wanted to validate our clustering analysis with similar parameters to check for different groupings through different methods. For this, we ran a K-Means clustering model and conducted the analysis using similar parameters derived from Recursive Feature Elimination with Cross-Validation (RFECV) with discrete parameters like drivetrain converted into a compatible and numerical classification variable.

These tasks are unsupervised machine learning, specifically clustering analysis. No target variable was provided, and the goal was to understand the hidden groupings with EVs based on their technical specifications. We chose GMM because it assumes that the data points were generated from a mixture of Gaussian distributions. This allows the clusters to have different variances, sizes, and shapes. GMM also allows for more flexibility for modeling continuous variables like speed, torque and efficiency which were not evenly distributed. We also chose K-Means as a sanity check for cluster choices and seeing if there were any big leaps and changes in characteristics or information to infer from another clustering method.

Findings

The lowest AIC value (−1222.13) occurred at 20 clusters, while the lowest BIC value (1304.32) occurred at 12 clusters, suggesting a trade-off between complexity and simplicity. The clustering revealed that high-performance brands such as Tesla, Porsche, Lucid, and Mercedes-Benz formed distinct clusters, while economy-focused models grouped together. This indicates that battery capacity and configuration are key differentiators in EV design, effectively categorizing vehicles into economy, mid-range, and premium tiers.

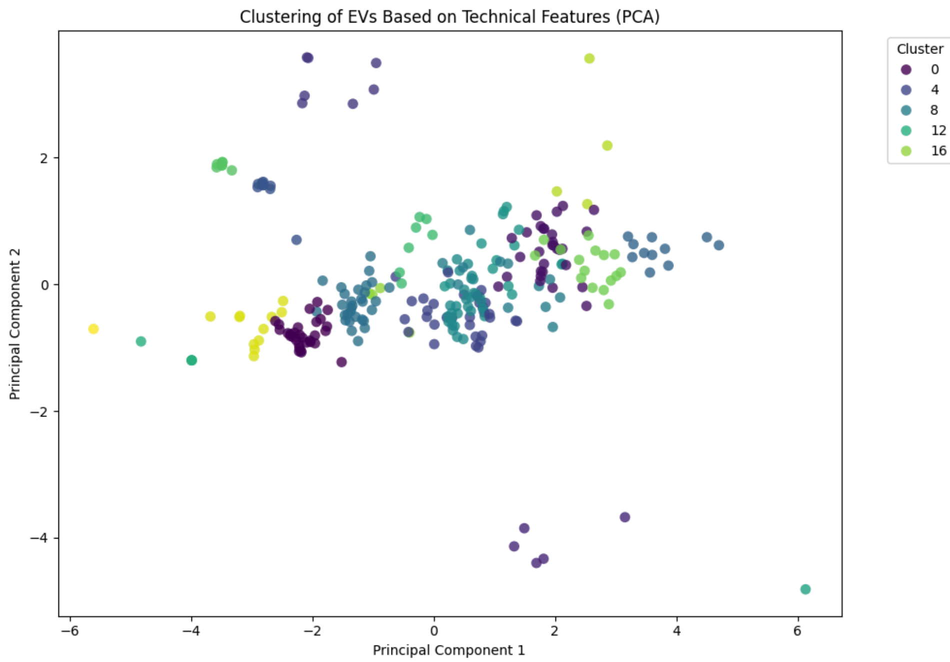

Figure 1. Clustering of EVs Based on Technical Features (PCA)

The PCA visualization shows how EVs are grouped based on their technical characteristics. Each point represents an individual EV model, and colors indicate the GMM cluster assignments. While some clusters form tighter groups, others overlap, suggesting that many models share similar performance profiles. The plot reinforces the idea that EV specifications exist along a continuous spectrum rather than forming discrete categories.

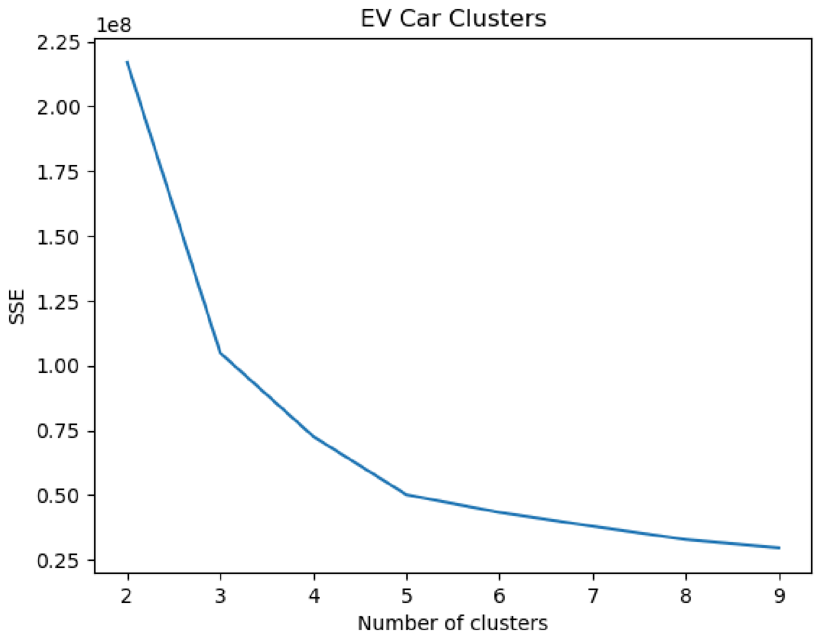

Figure 2. EV Car Clusters Based on SSE (Elbow Method)

Continuing off of the GMM clusterings, with the K-Means approach, we find that we are left with an optimal clustering of three rather than high number clusters ranging from ten to twenty. In figure 2, we can see that through SSE as the evaluator, there is the steepest drop at three with slightly less changes onward till five. Given this visualization, we went ahead and chose three clusters as our modeling approach (Figure 3).

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Given a total of 3 clusters through Kmeans

Cluster 1 (27 brands):
Abarth, Alfa, Alpine, Audi, BYD, CUPRA, Citroen, DS, Dacia, Fiat, Ford, Hyundai, Jaguar, Jeep, Kia, Lancia, Lexus, MG, Merced
es-Benz, Opel, Peugeot, Porsche, Renault, Skoda, Subaru, Toyota, Volkswagen

Cluster 2 (2 brands):
Lucid, Tesla

Cluster 3 (18 brands):
Audi, BYD, CUPRA, Cadillac, Ford, Genesis, Hyundai, Kia, Mercedes-Benz, NIO, Opel, Peugeot, Polestar, Porsche, Skoda, Volkswa
gen, Volvo, Zeekr
```

Figure 3. EV Car Clusters based on K-Means

Through the three clusters, as we can see in the above figure, there are many interesting findings from the brand names. In cluster two, we evidently see our most prominent figures in the EV space, Tesla and Lucid. Both of these brands manufacture present as high performance EVs that have stood the test of time and started out as EV forward car brands. Additionally, with the three clusters, we can see the distribution of brands with relation to their brand reputation (economy, mid tier, and premium models). Looking at the clusters, one seems to be the highest density of “economy” brands, while cluster three and two are associated with a higher tier branding. Another interesting finding is that brands like Polestar and Volvo, while sister companies, have very different reputations when it comes to building EVs. Polestar was started as an EV-only manufacturer while Volvo mainly focuses on traditional gas cars. However, we can see that they are clustered together, indicating that through technical specifications, they are considered relatively equivalent. While the cluster amount is different from the GMM approach, we can see again that parameters such as battery capacity and physical configurations helped make this determination.

Discussion

The GMM analysis showed that electric vehicles could be grouped into distinct groups based on how well they worked and how efficient they were. The differences between AIC and BIC show the trade-off between model complexity and simplicity. AIC (20 clusters), it finds more specific differences between models, while BIC (12 clusters) makes broader, more general groupings. The K-Means clustering found that there are optimally three clusters via the elbow method. While there could be an argument made for choosing a higher cluster count as SSE decreases, the biggest drop occurs between two to three clusters, which motivated us to make that decision. Additionally, for clarity and easier interpretation, a lower cluster count aided in our research in looking for how technical specifications can factor in a car brand's class (economy or premium). The end results show that EVs can be naturally divided into performance tiers based on how fast they can go, how efficient they are, and how much battery they have. The clustering was based only on numbers, so it gives an unbiased view of how to group vehicles without using brand or class labels.

3.2 Regression Analysis: Factors Influencing EV Efficiency

Methodology

For this analysis, the dependent variable was *efficiency_wh_per_km.* Multiple different variables were used as predictors including top speed, battery capacity, number of cells, torque, range, body dimensions, acceleration, seat count, towing capacity, and DC fast-charging power. Recursive Feature Elimination with Cross-Validation (RFECV) using a `LinearRegression()` estimator and five-fold cross-validation identified the optimal subset of 12 predictors.

The result showed us the best number of features, the best cross-validation score, and the features that were chosen. Using statsmodels, these chosen predictors were then put into an Ordinary Least Squares (OLS) regression model to get coefficients, p-values, and R^2 values.

Later, the model was run again with the drivetrain encoded numerically to see if the configuration of the drivetrain had a big effect on efficiency. Lastly, we used a logistic regression classifier to guess the type of drivetrain based on the same technical variables. Over 1,000 random splits, the model was trained and tested many times, and the average accuracy and confusion matrix for training and testing were printed.

This is a supervised learning problem that focuses on regression and classification. The OLS regression models the linear relationship between predictors and target variables with estimating coefficients that quantify each predictor's contribution to energy consumption. RFECV was used as a feature selection method to reduce weak or unnecessary predictors. It was used to make sure that only the most informative predictors were used and retained. The logistic regression model was used for the categorical prediction, which classified drivetrain type based on the vehicle specifications.

Findings

The RFECV model achieved a mean cross-validation R^2 of 0.60 and selected 12 key features. To better understand how the key technical variables relate to one another, a correlation heatmap was generated (Figure 4).

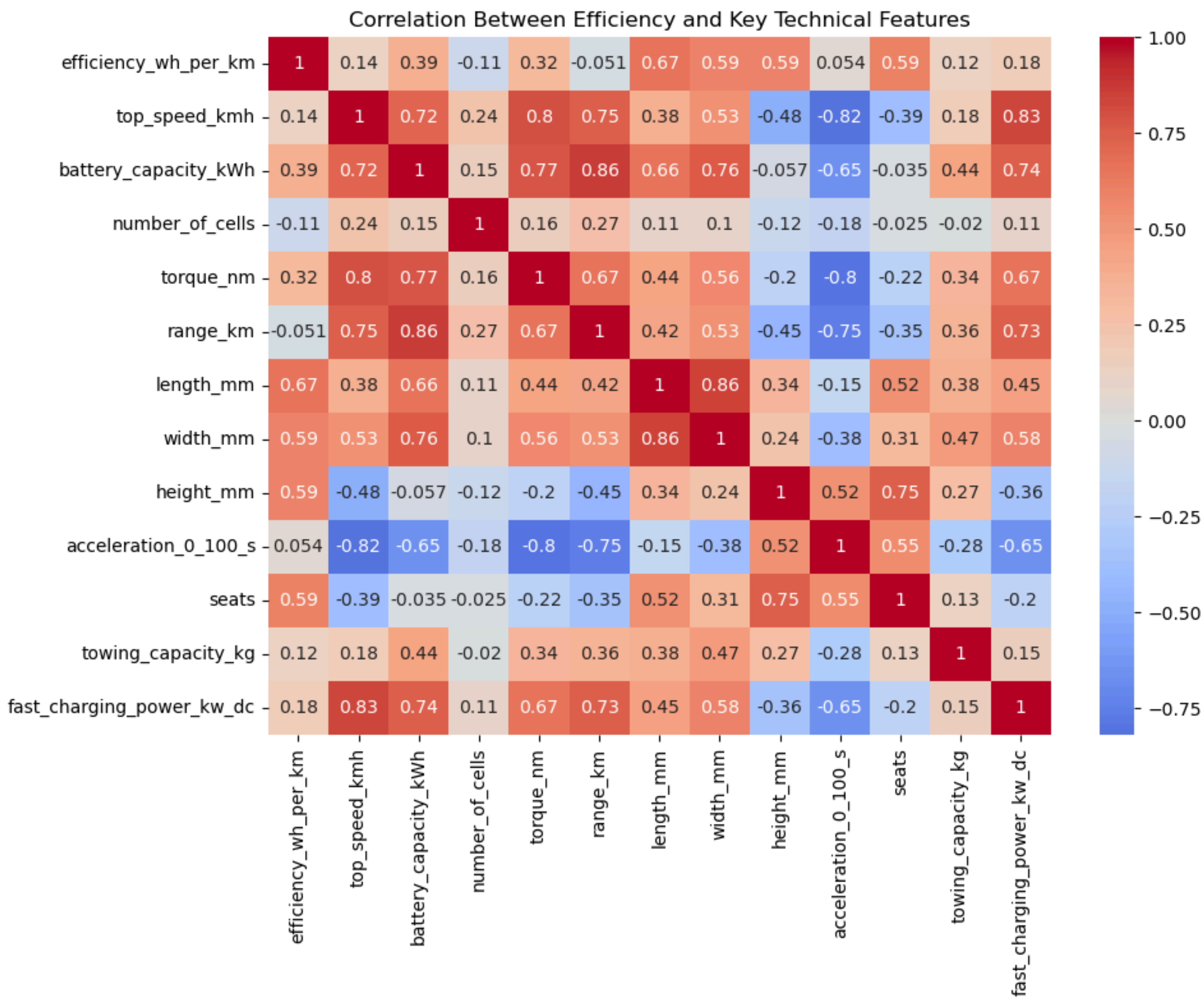

Figure 4. Correlation between efficiency (Wh/km) and key technical features of electric vehicles.

The heatmap shows that efficiency has moderate positive correlations with vehicle dimensions (length and height) and battery capacity, which suggests that larger and more powerful EVs consume more energy per kilometer. Strong correlations among battery capacity, range, and top speed confirm that these variables scale together, which contributes to multicollinearity in the regression model.

Following this, the OLS model had a R^2 of about 0.83, meaning the chosen features could explain about 83% of the changes in efficiency. Battery capacity, range, length, height, towing capacity, and top speed were statistically significant predictors; however, width was not. A high condition number ($\sim 2.86 \times 10^5$) (Cond. No. 2.86e+05) showed that there was multicollinearity among the predictors. This means that some features such as range and battery size, were related to each other. When the drivetrain was added, its p-value (~0.83) showed that it didn't have a big effect on efficiency, and the R^2 value ($R^2 = 0.834$) stayed about the same.

The average training accuracy for the logistic regression model that predicted drivetrain type was approximately 94%, while the average test accuracy was about 90% over 1,000 iterations. The confusion matrix showed that most of the predictions were right, but there were a few mistakes in the RWD and AWD categories.

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Confusion Matrix:
[[12  1  0]
 [ 2 12  3]
 [ 0  1 22]]
```

Discussion

This analysis shows that EVs efficiencies are primarily driven by the size, battery characteristics, and power-related features, instead of drivetrain configuration. The larger, heavier, and more powerful vehicles tend to consume more wattage per hour per kilometer. On the other hand, cars with longer ranges and optimized aerodynamics achieve higher efficiency. Unlike in gas/combustion vehicles, there is a statistical insignificance of drivetrain that suggest the efficiency of EVs depends less on whether the vehicle is Front, Rear, or All wheel drive and depends more on the total power output and aerodynamic design.

The high R^2 means that the linear model shows the efficiency well. However, the large conditional number shows overlapping relationships between certain technical features. The logistic model's high accuracy reinforces that the drivetrain configuration can still be interpreted from performance and design variables, specifically torque, acceleration, fast-charging capacity which were identified as the most predictive features. These key findings provide insight into engineering factors that determine EV energy consumption and suggest potential future efficiency improvements, like focusing on optimizing the vehicle size, weight, and power systems rather than drivetrain.

3.3 Predictive Modeling: Estimating EV Range

Methodology

Predictive analysis was conducted with *range_km* as the dependent variable. Eleven technical features, including battery capacity, torque, and acceleration were used as predictors in a multiple linear regression model. Linear assumption was evaluated to compare between predicted and actual range values using the residuals to observe how well the regression model performed.

Findings

The regression model achieved a cross-validation score of 0.93 and an R^2 of approximately 0.96, indicating strong predictive performance. Most predictors were statistically significant except *top_speed_kmh*, *width_mm*, and *fast_charging_power_kw_dc*. A high condition number (2.33×10^5) indicated multicollinearity among predictors, but the overall model fit remained excellent. This can be supported by comparing the actual and predicted values of the range (km) using a scatter plot (Figure 5). The predicted values mostly fit to the best fit line indicating a strong relationship between the model's predictions and the true values. This further suggests that the multiple linear regression model is effectively capturing the underlying relationships between the independent variables and the dependent variable (range (km)). The tight clustering around the line implies high predictive accuracy within the dataset further supporting the $R^2 \approx 0.96$.

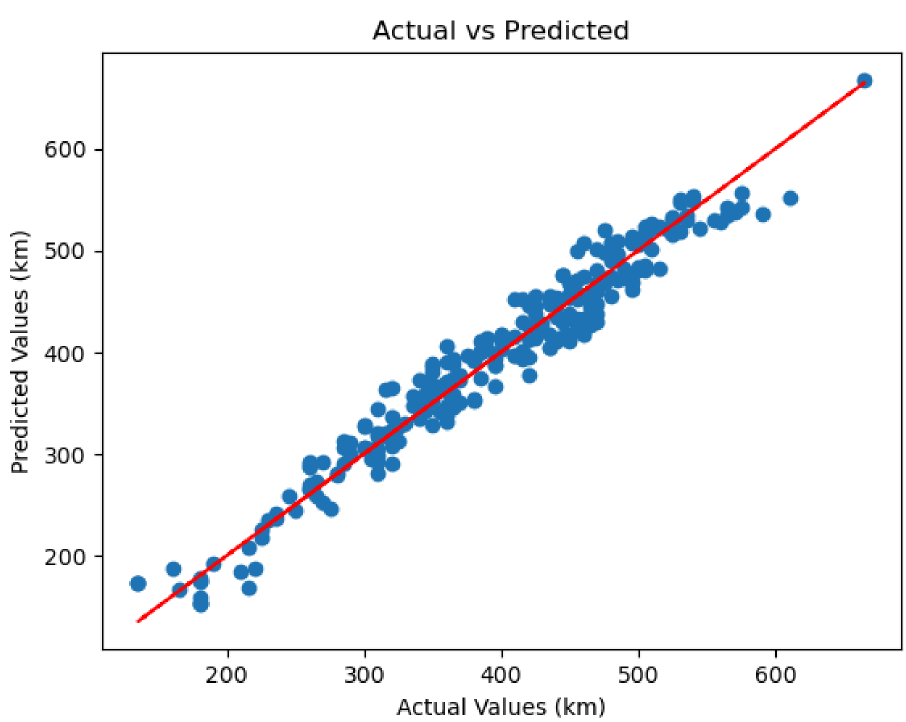

Figure 5: Scatter plot comparing the actual and predicted range (km) values from the multiple linear regression model.

Discussion

This model effectively predicts EV range using technical attributes. Key predictors such as battery capacity, number of cells, and torque were highly influential, suggesting that range can be reliably estimated from physical and power-related characteristics. The results provide practical value for both consumers and manufacturers, offering a data-driven approach to understanding and optimizing EV performance.

Discussion

Across all analyses, the results consistently highlight the central role of battery-related variables in defining EV performance and efficiency. Clustering identified clear performance tiers across brands, regression analysis revealed which features drive efficiency, and predictive modeling accurately estimated range. Together, these models show that EV efficiency depends more on energy storage and aerodynamic design than on drivetrain configuration. The integration of unsupervised and supervised learning demonstrates the potential of machine learning to reveal actionable insights in sustainable vehicle design.

While the models achieved strong results, it is important to acknowledge that possible limitations and biases exist. The dataset could overrepresent certain manufacturers or model types, such as high-end EVs; this has the potential to bias the results toward performance-oriented designs rather than mass marketed vehicles. Additionally, efficiency values might not reflect real-world driving conditions because they are based on standardized test cycles. Further analysis could be done by incorporating real world energy consumption data and environmental variables.

Conclusion

This study used machine learning methods to analyze 478 electric vehicle models and identify the technical factors that most influence efficiency and range. Through GMM clustering, linear regression, and predictive modeling, we found that EV performance is largely shaped by battery capacity, range, and power output, while drivetrain configuration plays a limited role. The models achieved strong predictive performance (R^2 up to 0.96), confirming that EV range and efficiency can be reliably estimated using measurable specifications. These findings contribute to understanding the design trade-offs in electric vehicle engineering and demonstrate how data-driven approaches can guide innovation toward more efficient, sustainable transportation.

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